Auxiliary-input leads 2026-11-12

Auxiliary inputs, curves and points: lead chase and five follow-ups (2026-11-12)

Question. What is the current state of every lead about auxiliary curves, auxiliary points and Cheon's DLP with auxiliary inputs, and how far can the five follow-ups the owner approved on 2026-11-12 be taken from this session? **Class.** Desk work and literature only. Nothing was run, launched or measured. Every statement is `[pcc:unverified]`. **Report PDF:** `report.pdf`, built from this file. **Figures:** `figures/lead_map.svg` (lead state) and the recorded exponent graph of the wide study, `../cheon_ecbench_wide_20261106/alpha.svg`.

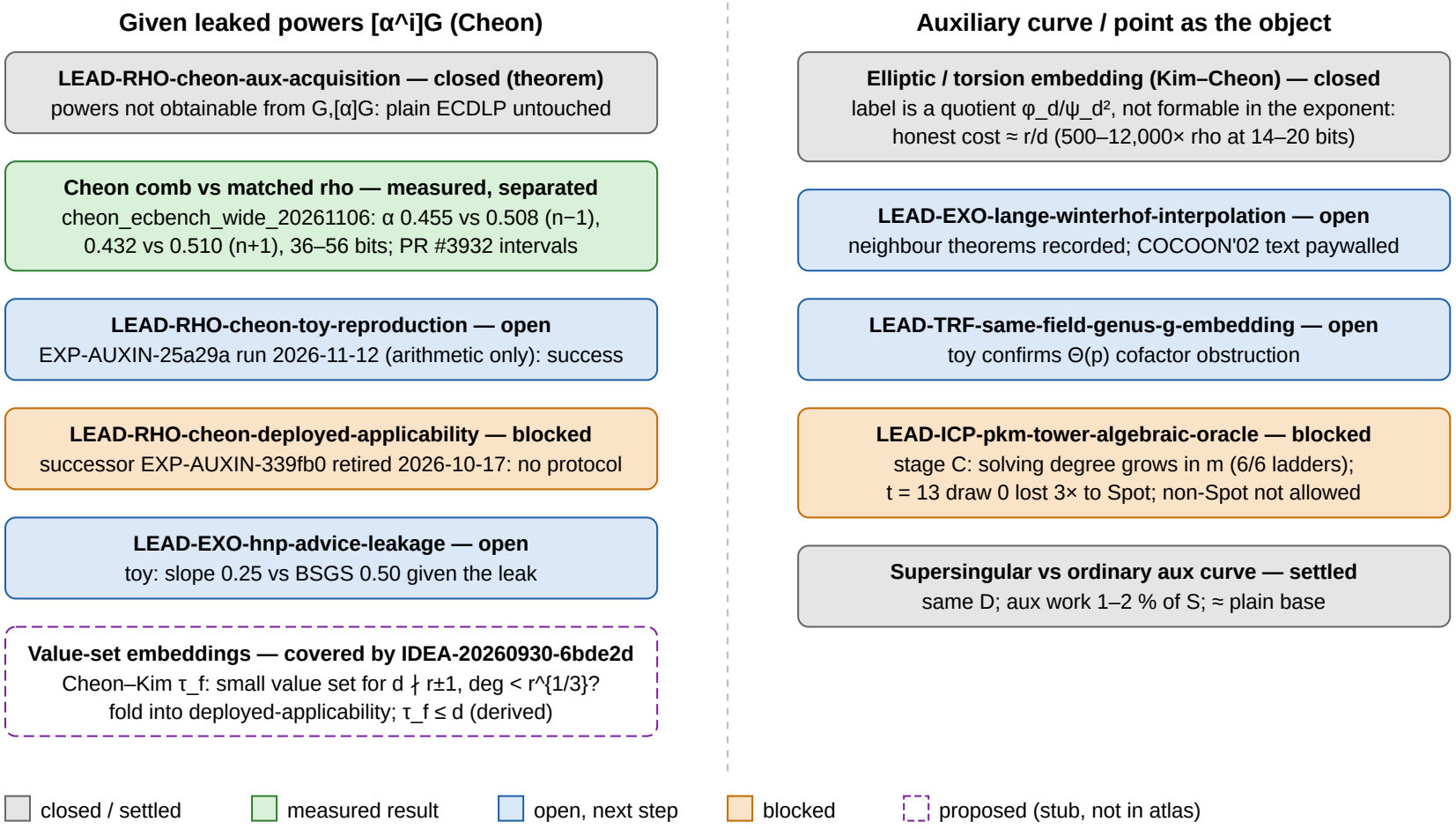
Sources checked

- research-kb: **unavailable** (no MCP server connected; `scripts/research-kb` finds no catalog in this container).
- Lead atlas: `crypto-autoresearcher` cloned at `8c82ddc8` (2026-11-11). Every atlas file is AES-SIV vault ciphertext (`VLT1`); `KEY` is not set, so `tools/lead_atlas.py` cannot run. Lead fields were read from the in-repo copies: `research/lead_sweep_20261110/leads.json` and per-lead `RESULT.md`, `research/prime_field_lead_assessment_20261107/atlas_survey.md`, `papers/ecdlp-index-calculus-treatise/survey/generic.md` §19.
- **Update 2026-11-12:** the vault was unlocked later the same day (`vaultctl.py setup` + `unlock`, 27,507 files in the sparse checkout). Read: `knowledge/leads/ecdlp/generic/LEAD-RHO-cheon-{toy-reproduction,deployed-applicability}.yaml`, `ledger/hypotheses/H-AUXIN-{a86953,6db354}.yaml`, `experiments/EXP-AUXIN-25a29a/specification.yaml`, `experiments/EXP-AUXIN-7e2e3d/amendments/`, `ledger/decisions/DEC-20260923-a4c3d5`, `DEC-20260930-{224fc1,8af20f,ca32c7}`, `DEC-20261016-5bdc04`, `DEC-20261017-0fab25`, `ledger/proposals/IDEA-20260930-6bde2d.yaml`; full-text search of the decrypted `knowledge/` and `ledger/{hypotheses,proposals}` for Cheon–Kim / value set / τ_f / 2012/609.
- Studies: `research/cheon_ecbench_wide_20261106/`, `cheon_ecbench_medium_20261106/`, `cheon_p224_shape_20261022/`, `torsion_auxiliary_inputs_20260918/`, `srs_auxiliary_census_20260927/`, `pkm_supersingular_aux_20261015/`, `pkm_isogeny_tower_p192_20261015/`, `pkm_matched_prime_20261106/`, `pkm_falldegree_unblock_20261107/`; `research/notes/ecdlp-general/RESEARCH_TORSION_AUXILIARY_INPUTS.md`.
- Open PRs: #3932 (pooled Cheon intervals), #3938 (13 new leads).
- Literature: Kiltz–Winterhof, *Bull. Austral. Math. Soc.* 69 (2004) 305–315 (read in full); Lange–Winterhof COCOON'02 and AAECC'03 (metadata only, paywalled). See `LITERATURE_LANGE_WINTERHOF.md`.

1. Lead state

Auxiliary inputs, curves and points: lead state on 2026-11-12

every status [pcc:unverified]; atlas read from the unlocked vault, 2026-11-12



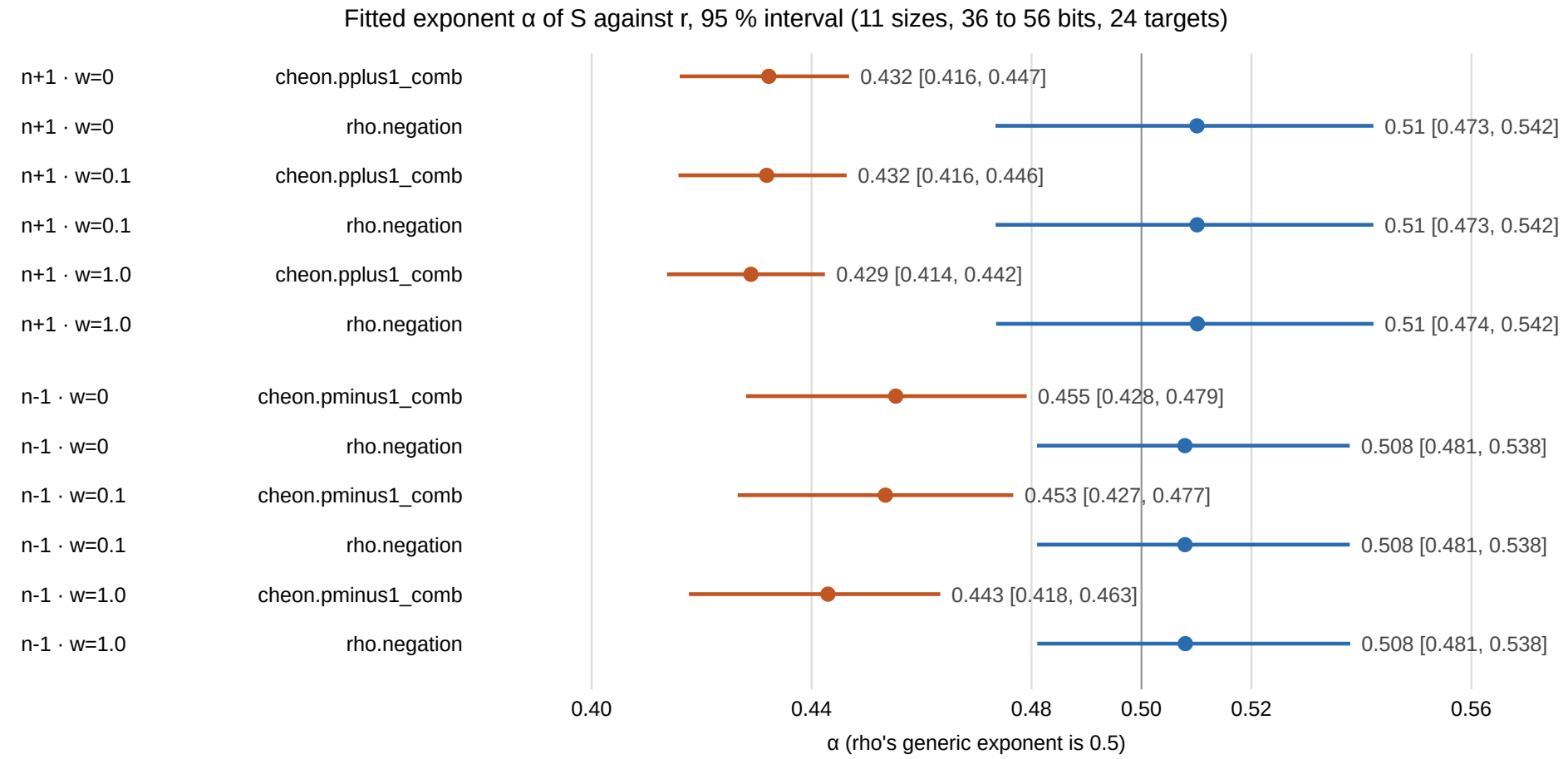
Lead map

lead	disposition	latest evidence
`LEAD-RHO-cheon-aux-acquisition`	closed_obstructed (theorem)	`[α²]G` from `G, [α]G` is square-DH. The 2026-11-10 sweep toy was inconclusive (positive control and A/A failed).
Cheon comb vs matched rho, powers given	measured	Wide study: α 0.455 [0.428, 0.479] vs rho 0.508 [0.481, 0.538] (n−1), 0.432 [0.416, 0.447] vs 0.510 [0.473, 0.542] (n+1), w = 0, 36–56 bits, 1,056 replayed runs. Pooled ratio comb/rho 0.525 [0.473, 0.584] (n−1), 0.367 [0.318, 0.422] (n+1) in open PR #3932.

lead	disposition	latest evidence
LEAD-RHO-cheon-toy-reproduction	open_next_step	Sweep toy matched 1.5(√(r/d)+√d) within 4 %; inconclusive because one negative-control cell sat at exactly 5 %. EXP-AUXIN-25a29a (arithmetic only) run 2026-11-12: success (§2, item 1).
LEAD-RHO-cheon-deployed-applicability	open_blocked	No current protocol: successor EXP-AUXIN-339fb0 retired 2026-10-17 (§2, item 4). Powers-of-tau census: 5.74–14.32 bits.
LEAD-EXO-hnp-advice-leakage	open_next_step	Sweep toy: Cheon slope 0.2516 vs BSGS 0.5039 given the leak; its rho baseline is mis-specified (slope 1.006), so its rho ratios are void.
Elliptic / torsion embedding (Kim–Cheon)	closed by measurement; no atlas lead	Label is a quotient; honest cost ≈ r/d.
LEAD-EXO-lange-winterhof-interpolation	open_next_step	Neighbour theorems recorded (§2, item 3).
LEAD-TRF-same-field-genus-g-embedding	open_next_step	Sweep toy confirms #Jac = #E₁·#E₂ on 132 curves, the Θ(p) cofactor obstruction.
LEAD-ICP-pkm-tower-algebraic-oracle	open_blocked	Stage C: solving degree grows in m on 6 of 6 ladders. Matched primes: D follows t, not p, and equals the null. t = 13 undecided (§2, item 5).
Supersingular vs ordinary auxiliary curve	settled	Same D in 14 cells; auxiliary work 0.9–1.9 % of S vs 10–27 % searched; against a plain base unresolved or 1–7 % dearer.
Cheon–Kim value-set embeddings	covered by IDEA-20260930-6bde2d (proposed)	Ceiling gain ≤ √q_max for the whole polynomial family (§2, item 2).

A conflict resolved. The 2026-09-18 note's Cheon exponent 0.23 and the ecbench exponents of about 0.45 are both consistent with theory: the first used d ≈ √r (predicted 1/4), the ecbench studies use P-224-shaped d ≈ r^0.129 and r^0.211 (predicted (1 − ρ)/2 ≈ 0.435 and 0.394). Different d, not a contradiction.

A retraction to carry. The Cheon/rho ratios of research/cheon_p224_shape_20261022/ were withdrawn on 2026-10-23 (H3 gate failed); the wide study supersedes them.



Fitted exponents, wide study (recorded values, 95 % intervals)

2. The five follow-ups

item	outcome	why / what was done	what unblocks it
1. Run EXP-AUXIN-25a29a (H-AUXIN-a86953: 56-bit r, d = 65536, stage counts, gain fractions)	**run, success** (update 2026-11-12, vault unlocked)	The frozen specification is integer arithmetic only: no curve, no walk, two routes, 1 GB. Both routes agree on every preregistered value: `exp_auxin_25a29a/RESULT.md`. Executed here, not in the ledger's run package (no write access to `crypto-autoresearcher`).	A session with write access mints the RUN id and files the package. A memory-charged 56-bit walk is a separate, not yet designed experiment.
2. File a lead for value-set / auxiliary embeddings	**not a new lead** (update 2026-11-12)	The vault shows prior art: IDEA-20260930-6bde2d (proposed, linked from `LEAD-RH0-cheon-deployed-applicability`) prices the whole polynomial-embedding family, Kim–Cheon general `f` included, with ceiling gain ≤ √d ≤ √q_max from τ_f ≤	Fold into the existing lead's notes, not a new id.

item	outcome	why / what was done	what unblocks it
		deg f. The stub's existence question is a sub-question of that lead. See `LEAD_STUB_value_set_embedding.md`.	
3. Lange–Winterhof exact theorem	**partly done**	Kiltz–Winterhof 2004 Theorems 5 and 6, stated as extending Lange–Winterhof Theorem 3, are transcribed with scope in `LITERATURE_LANGE_WINTERHOF.md`. The COCOON'02 text is still paywalled.	Library access or an author copy.
4. Approve the revised deployed-applicability protocol	**nothing to approve** (update 2026-11-12)	The vault shows the chain: amendment d1d10 refused (DEC-20260923-a4c3d5); revisions typed, narrow and gated (2026-09-26 to 09-28); successor contract EXP-AUXIN-339fb0 approved at sha256 `6a1dca32…` (DEC-20260930-ca32c7) and authorized (DEC-20260930-224fc1), never run, then **retired** on 2026-10-17 by DEC-20261017-0fab25 as "a pre-schema draft… never an approved experiment contract". H-AUXIN-6db354 is still `proposed`. The atlas's next step ("approve a revised protocol") points at no current document.	A new protocol under H-AUXIN-6db354 must be written and reviewed in `crypto-autoresearcher`; only then can it be approved.
5. Re-launch PKM `pfd-x-t13-k0-c6` on non-Spot capacity	**blocked by policy**	`taskq/deploy/eks-karpenter/spot-only-policy.yaml` admits only Spot NodePools; `COST_CONTROLS.md` forbids an On-Demand fallback; the CRIU resumable route is qualified to an 8 GiB envelope (the cell needs 56 GiB); this container has 15 GB. Three Spot attempts were already lost (`SPOT_LOSSES.md`).	An administrative exception for one On-Demand node (policy change), or an owner decision for a fourth Spot attempt, or a memory-qualified CRIU route.

3. Next steps

1. Done 2026-11-12 with the vault key: item 1 run, items 2 and 4 re-read (above). Remaining in `crypto-autoresearcher` (needs write access): file the EXP-AUXIN-25a29a run package; correct both Cheon leads' next-step text; write a new census protocol under H-AUXIN-6db354.
2. Decide item 5: a one-off On-Demand exception, a fourth Spot attempt, or accept "undecided at t = 13".
3. Read Cheon–Kim 2016 §§3–5 for the value-set stub's next step (desk).